

Project Title
Hospital Bed / OR Scheduling

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Hospitals face a complex scheduling challenge: assigning patients to beds and operating rooms while balancing procedure durations, patient priorities, staff availability, and deadlines. Inefficient scheduling leads to idle resources, longer waits, and compromised outcomes.

This project develops an optimization engine that intelligently allocates patients and procedures to rooms and time slots, aiming to minimise makespan and idle time while respecting all constraints. The system integrates algorithms for resource utilisation, priority handling, and deadline adherence, producing schedules that improve throughput and patient care.

A reporting module ensures transparency, offering insights into utilisation rates, bottlenecks, and quality metrics. The engine is designed to adapt dynamically to emergencies, cancellations, or staff changes, ensuring resilience in real-world hospital operations. By combining operational efficiency with patient-centric scheduling, the solution directly enhances hospital capacity management, staff productivity, and clinical outcomes.

Signature of the Guide